COMP90050 ADBS

Week 9

Q1 Concurrency

1. Assume the following two transactions start at nearly the same time and there is no other concurrent transaction. The 2nd operation of both transactions is Xlock(B). Is there a potential problem if Transaction 1 performs the operation first? What if Transaction 2 performs the operation first?

Transaction 1		Transaction 2	
1.1	Slock(A)	2.1	Slock(A)
1.2	Xlock(B)	2.2	Xlock(B)
1.3	Read(A)	2.3	Write(B)
1.4	Read(B)	2.4	Unlock(B)
1.5	Write(B)	2.5	Read(A)
1.6	Unlock(A)	2.6	Xlock(B)
1.7	Unlock(B)	2.7	Write(B)
		2.8	Unlock(A)
		2.9	Unlock(B)

T1 first: No problem, as T1 releases the the lock at the end and T2 has to wait till then.

T2 first:

- T1 may have a dirty read of B.
- When T2 releases the lock (2.4), T1 will be granted the Xlock on B (1.2) and perform its read/write. Once T1 completes, T2 will acquire another Xlock on B (2.6) and modify the object.
- I.e. reading of B by T1 happens between the two writes of B by T2.

Q2 Degree of Isolation

2. What degree of isolation does the following transaction provide?

Slock(A)
Xlock(B)
Read(A)
Write(B)
Read(C)
Unlock(A)
Unlock(B)

Degree 3: two phase and well formed
Degree 2:
- two phase with respect to exclusive locks
- well formed with respect to reads and writes

Degree 1:
- two phase with respect to exclusive locks
- well formed with respect to writes

Degree 0: well-formed with respect to writes

Degree 1 provided.
- There is one read operation 'Read(C)' without taking any lock.
- The only write operation has an exclusive lock associated with it.
- The transaction is two-phase with respect to exclusive lock.

Q3 Degree of Isolation

3. The following operations are given with Degree 2 isolation locking principles in place. Convert the locking sequence to Degree 3.

Degree 2
Slock(A)
Read(A)
Unlock(A)
Xlock(C)
Xlock(B)
Write(B)
Slock(A)
Read(A)
Unlock(A)
Write(C)
Unlock(B)
Unlock(C)

Degree 3: two phase and well formed

Degree 2:
- two phase with respect to exclusive locks
- well formed with respect to reads and writes

Degree 1:
- two phase with respect to exclusive locks
- well formed with respect to writes

Degree 0: well-formed with respect to writes

Degree 3
Slock(A)
Read(A)
Xlock(C)
Xlock(B)
Write(B)
Read(A)
Unlock(A)
Write(C)
Unlock(B)
Unlock(C)

Lab Sheet

Week 9

FEIT Subject Surveys

- Sent to university email address with the subject "FEIT Subject Surveys”
- Open until Sunday 17 May